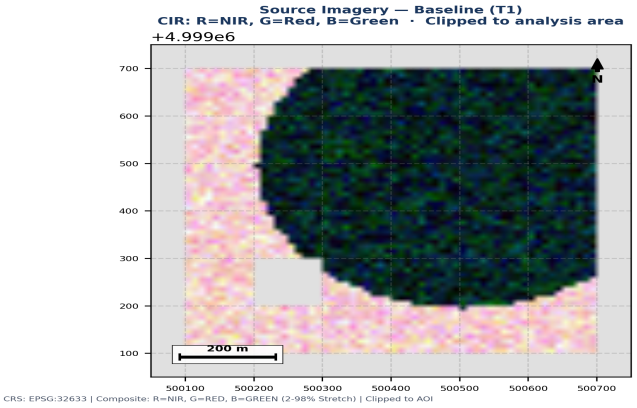
WETLAND DEGRADATION ASSESSMENT

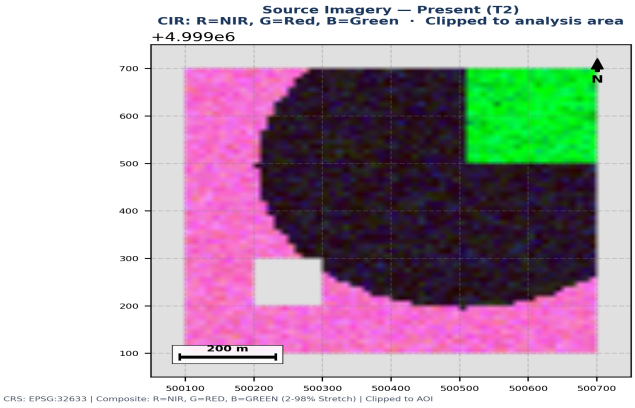
Assessment of ecological change within the selected analysis area

OBSERVATION PERIOD	2023-08-14 → 2023-08-28	INTERVAL	14 days
WETLAND CONTEXT	Project-provided wetland boundary	RUN ID	run_20260922_082400
ANALYSIS FOOTPRINT	100.00% (35.00 ha)	SPATIAL REFERENCE	EPSG:32633

SOURCE IMAGERY — ANALYSIS AREA OVERVIEW



T1 Baseline · 2023-08-14

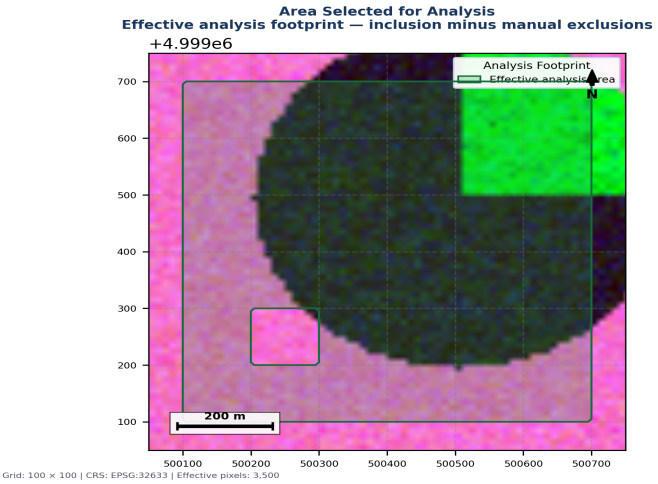


T2 Present · 2023-08-28

OVERALL WETLAND ASSESSMENT

NO MEASURABLE WETLAND DEGRADATION DETECTED

WETLAND-LEVEL INDICATOR	OBSERVED RESULT
Hydro-Ecological Degradation (primary)	0.00%
Hydro-Ecological Recovery	0.00%
Vegetation Degradation (supporting evidence)	0.00%
Vegetation Recovery (supporting evidence)	0.00%
Surface-Water Loss (Contraction)	0.00%
Surface-Water Gain (Expansion)	10.80%
Mean CVA Spectral Displacement	0.6969
Effective Analysis Area	100.00% (35.00 ha)



Analysis area footprint

Available multi-stream remote-sensing evidence demonstrates no systemic wetland degradation within the selected analysis area across the observation window.

1. THE AREA WE ANALYZED

This assessment examines changes within the area selected for analysis. Vegetation condition, spectral indicators and surface-water evidence are compared between the two observation dates. The selected area defines where the assessment is performed and should not automatically be interpreted as a legal or statutory wetland boundary.

The analysis is spatially bounded by the configured wetland boundary. All ecological condition indicators are evaluated within this defined area.

SOURCE IMAGERY — T1 vs T2 COMPARISON

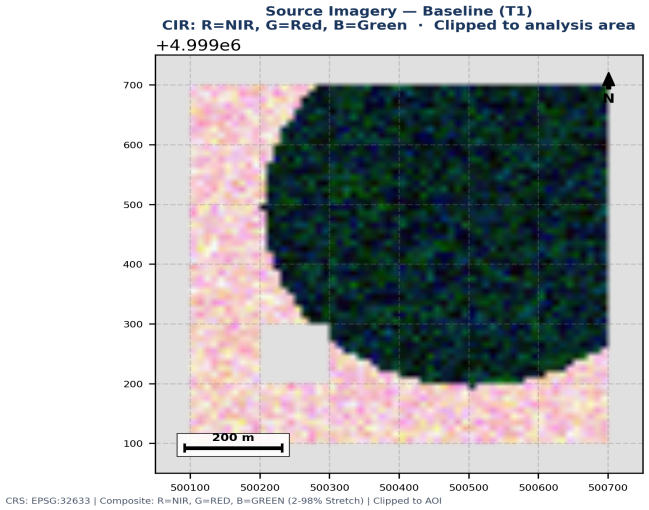


Figure 1a: Baseline imagery (T1, 2023-08-14) — restricted to effective analysis area

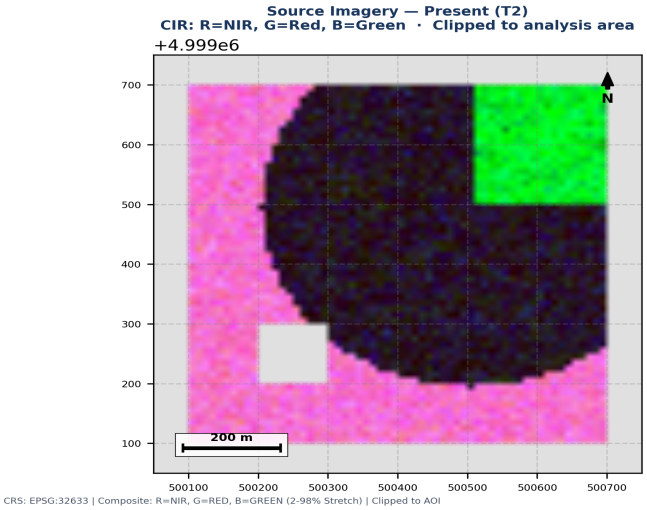
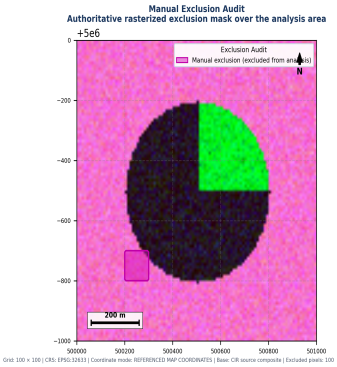


Figure 1b: Present imagery (T2, 2023-08-28) — same geographic extent

SPATIAL ATTRIBUTE	OBSERVED / CONFIGURED VALUE
Wetland context status	WETLAND_BOUNDARY_AVAILABLE
Wetland source	Project-provided wetland boundary
Source dataset / boundary	wetland_boundary.geojson
Selected analysis area	100.00% (36.00 ha)
Effective analysis footprint	100.00% (35.00 ha)
Manual exclusion footprint	2.78% (1.00 ha)
Physical wetland loss pathway	NOT_ASSESSED



Effective analysis footprint

2. WETLAND CONDITION

Wetland condition is evaluated by integrating multiple independent evidence streams. Vegetation condition, continuous spectral dynamics, and surface-water presence are each assessed across the active analysis footprint. The source imagery below shows the actual multispectral input at each observation date, clipped to the effective analysis area.

SOURCE IMAGERY — OBSERVATION OVERVIEW

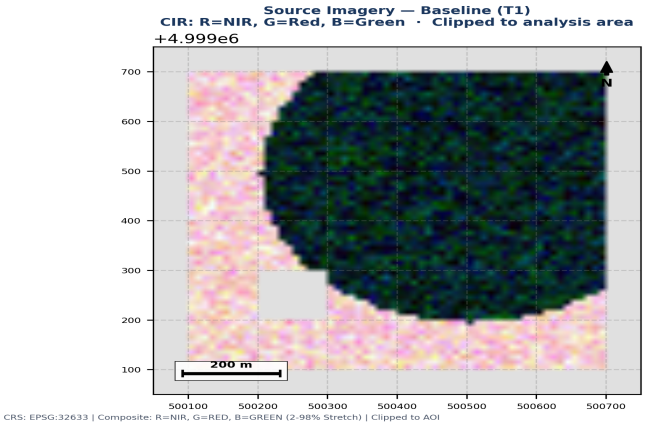


Figure 2a: Baseline observation (T1, 2023-08-14) — effective analysis area

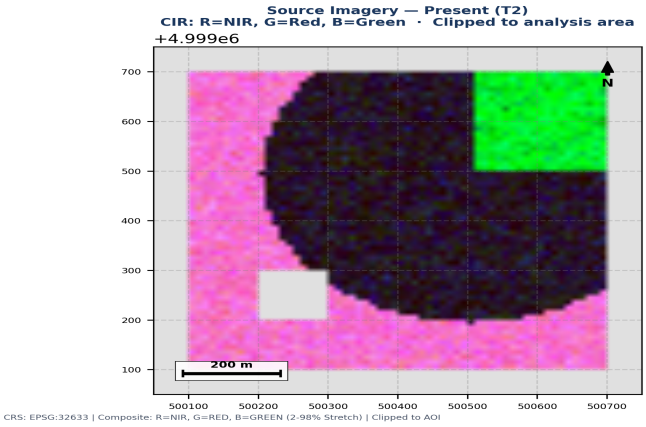


Figure 2b: Present observation (T2, 2023-08-28) — effective analysis area

EVIDENCE STREAM SUMMARY

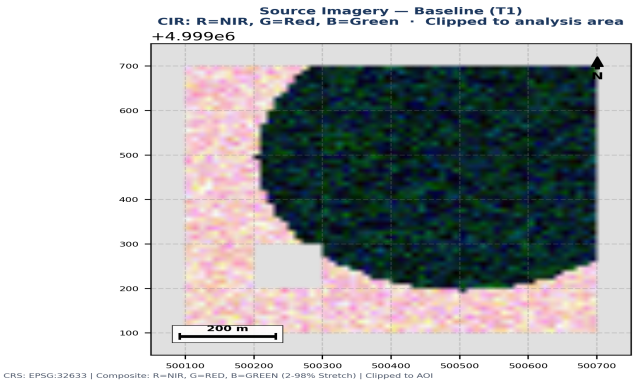
EVIDENCE STREAM	OBSERVED FINDING	ROLE IN WETLAND ASSESSMENT
Vegetation condition	Stable canopy state (100.00% persistent)	Tracks canopy structure and foliar condition class shifts across the analysis area.
Spectral change	Mean CVA: 0.6969 | ΔNDVI: +0.4944	Measures continuous sub-classification physiological canopy displacement.
Surface-water change	Loss: 0.00%; Gain: 10.80%	Independently evaluates wetting and drying dynamics.
Integrated assessment	**NO MEASURABLE WETLAND DEGRADATION DETECTED**	Synthesizes all evidence streams into the wetland degradation finding.

Optical vegetation stress without concurrent hydrological contraction indicates vegetation-level dynamics. Concurrent vegetation stress and water loss provide multi-stream evidence of wetland degradation.

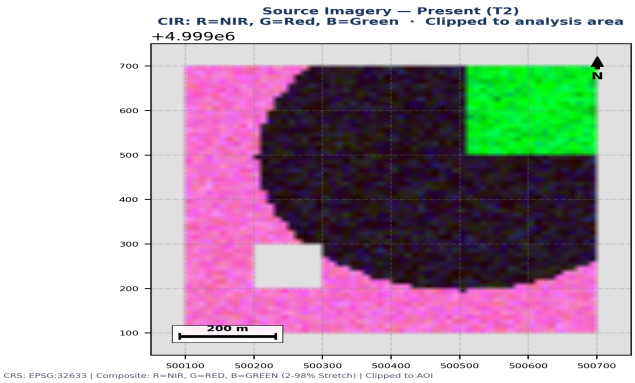
3. VEGETATION EVIDENCE

Vegetation condition serves as an essential line of evidence. Canopy condition is classified at baseline (T1) and present (T2) into four discrete states: Healthy Vegetation, Moderate Stress, Severe Stress, and Background / Non-Vegetated. All proportions are computed as percentages of the valid analysis area (classified footprint within the effective analysis boundary).

SOURCE IMAGERY CONTEXT



T1 — 2023-08-14



T2 — 2023-08-28

VEGETATION CONDITION CLASS DISTRIBUTION

CONDITION CLASS	BASELINE (T1) %	PRESENT (T2) %	NET CHANGE (%pts)
Healthy vegetation	0.00%	0.00%	0.00%
Moderate stress	0.00%	0.00%	0.00%
Severe stress	0.00%	0.00%	0.00%
Background / non-vegetated	100.00%	100.00%	0.00%

VEGETATION TRANSITION DYNAMICS

TRANSITION PATHWAY	INCLUDED TRANSITIONS	% OF EFFECTIVE FOOTPRINT
Persistent classified state	H→H, M→M, S→S, B→B	100.00%
Vegetation degradation pathways	H→M, H→S, M→S	0.00%
Vegetation recovery pathways	M→H, S→M, S→H	0.00%
Canopy loss / non-vegetated transition	H→B, M→B, S→B	0.00%
Canopy emergence / colonisation	B→H, B→M, B→S	0.00%

VEGETATION TRANSITION MAPS

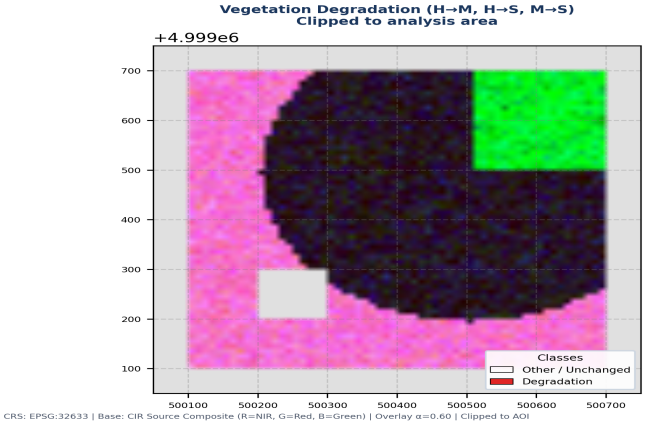


Figure 3a: Vegetation degradation pathways (H→M, H→S, M→S)

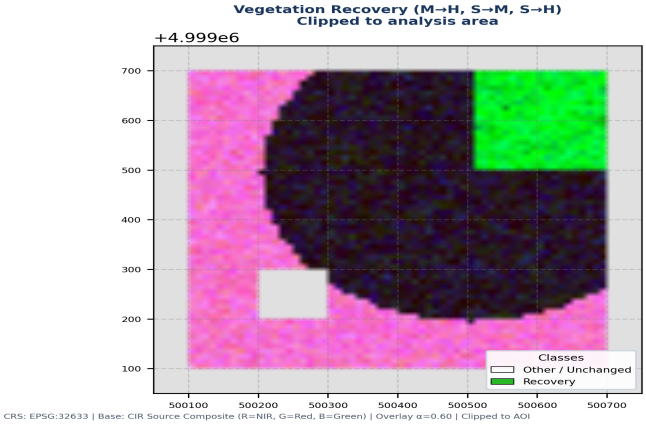


Figure 3b: Vegetation recovery pathways (M→H, S→M, S→H)

Vegetation evidence summary: 100.00% of the active analysis area remained in a stable classified condition state. No vegetation degradation or recovery pathways were detected.

4. SPECTRAL EVIDENCE

Continuous spectral indices provide independent evidence of foliar and physiological condition shifts. Change Vector Analysis (CVA) integrates ΔNDVI, ΔGNDVI, and ΔNDRE in a multi-dimensional feature space, capturing subtle canopy adjustments within or across categorical classification boundaries. These metrics represent continuous radiometric evidence rather than standalone wetland degradation measures.

SPECTRAL INDICATOR	MEAN	MEDIAN	P95	% POSITIVE
CVA Magnitude (‖ΔV‖)	0.6969	0.6719	1.3386	N/A
ΔNDVI	0.4944	0.2951	1.0850	87.1%
ΔGNDVI	0.2689	0.0051	0.8085	84.4%
ΔNDRE	-0.0334	-0.0046	0.3494	28.6%

SPECTRAL DYNAMICS & VEGETATION INDEX DIFFERENCE MAPS

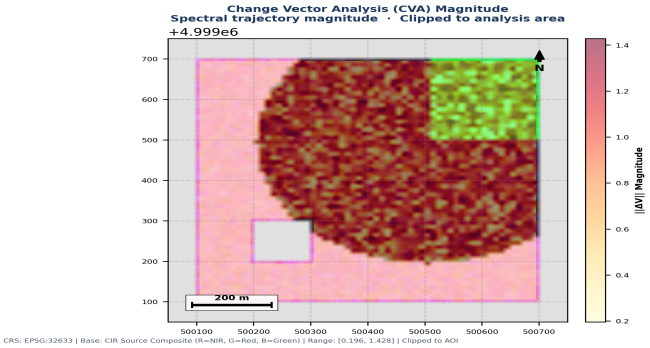


Figure 4a: CVA magnitude (radiometric displacement)

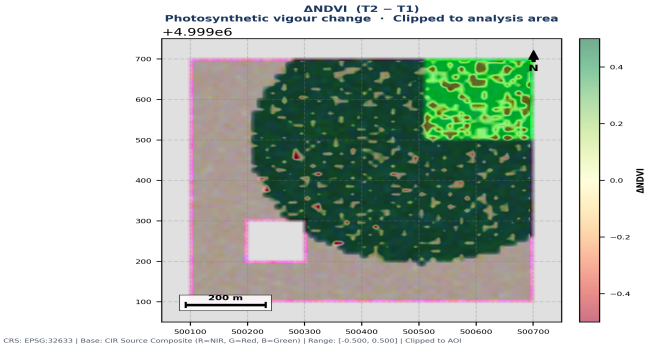


Figure 4b: ΔNDVI (vegetation greenness change)

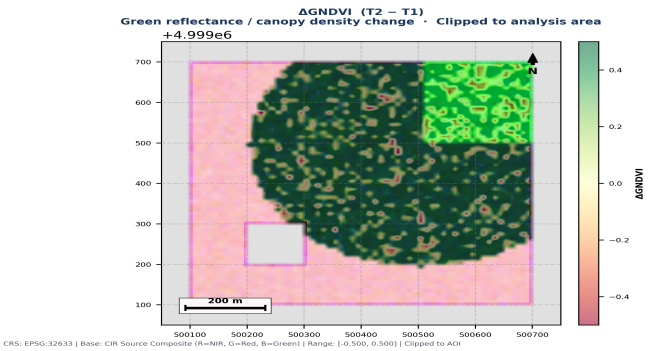


Figure 4c: ΔGNDVI (green vegetation index change)

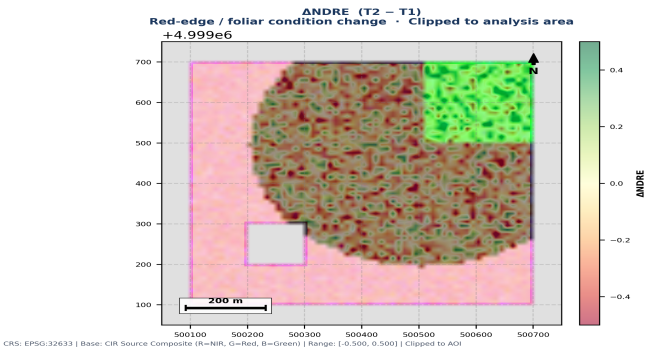


Figure 4d: ΔNDRE (red-edge canopy stress change)

Spectral evidence summary: Continuous radiometric monitoring records an average CVA displacement of 0.6969, reflecting sub-classification physiological variation across the vegetative canopy.

5. SURFACE-WATER EVIDENCE

Surface-water presence and moisture dynamics provide an essential hydrological line of evidence. Water presence is evaluated from the Normalized Difference Water Index (NDWI) using a calibrated threshold of 0.00. Source imagery is shown alongside NDWI maps to contextualise the hydrological assessment.

SOURCE IMAGERY CONTEXT — OBSERVATION DATES

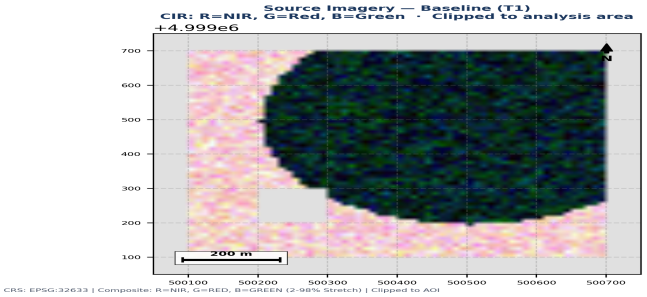


Figure 5a: Baseline imagery (T1, 2023-08-14)

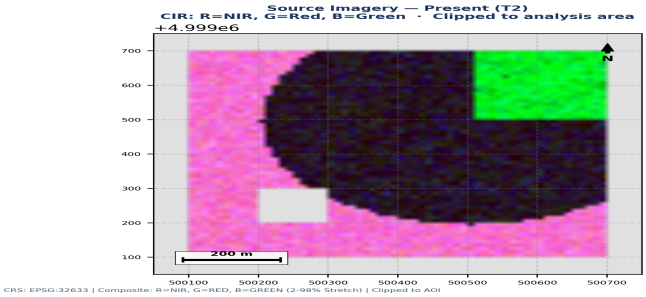


Figure 5b: Present imagery (T2, 2023-08-28)

NDWI SURFACE-WATER INDEX — T1 vs T2

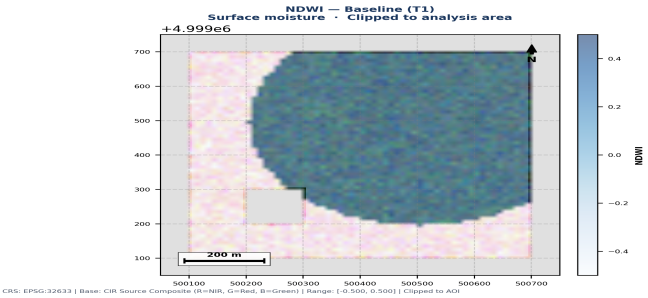


Figure 5c: Baseline NDWI (T1, 2023-08-14)

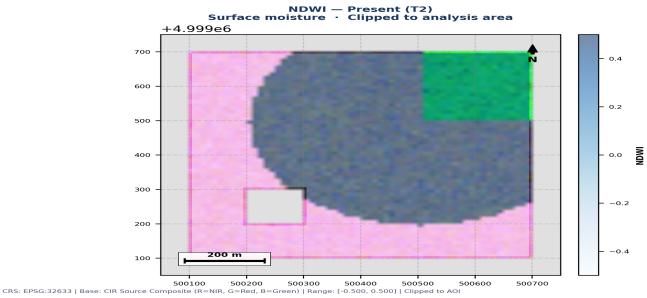


Figure 5d: Present NDWI (T2, 2023-08-28)

HYDROLOGICAL METRICS & SURFACE-WATER DYNAMICS

HYDROLOGICAL METRIC	% FOOTPRINT	PHYSICAL	DEFINITION
Baseline Water (T1)	0.77%	0.27 ha	Surface water detected at baseline
Present Water (T2)	11.57%	4.05 ha	Surface water detected at present
Water Expansion (Gain)	10.80%	3.78 ha	Newly emerged / wetting
Water Contraction (Loss)	0.00%	0.00 ha	Inundated becoming dry
NDWI Threshold	0.00	N/A	Calibrated NDWI zero-plane

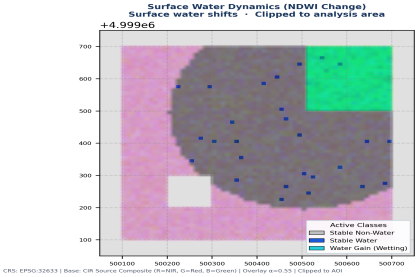


Figure 5e: Water change map

Hydrological evidence summary: Surface-water dynamics indicate 0.00% contraction (drying) and 10.80% expansion (wetting) across the analyzed area.

6. WETLAND DEGRADATION ASSESSMENT

This section presents the primary analytical synthesis. All available independent evidence streams — vegetation condition, continuous spectral shifts, and surface-water dynamics — are evaluated to determine whether wetland degradation occurred within the selected analysis area between observation dates.

Scientific distinction: Vegetation Degradation vs. Wetland Degradation

Vegetation stress alone does not establish wetland degradation. A hydro-ecological wetland degradation signal requires concurrent evidence of both vegetation deterioration and surface-water loss. Where surface-water evidence is absent or stable, vegetation shifts represent vegetation-only condition responses (physiological stress, seasonal variation) rather than systemic wetland loss.

INTEGRATED EVIDENCE MAP

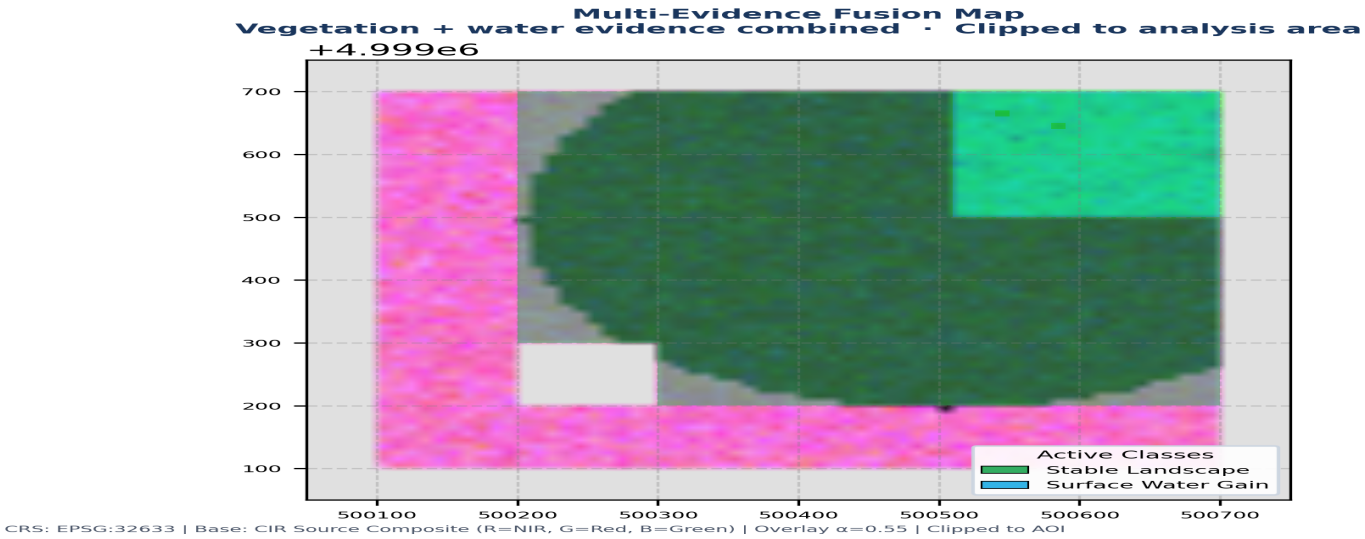


Figure 6a: Multi-evidence fusion map combining vegetation transitions and surface-water dynamics across the effective analysis footprint.

PRIMARY ASSESSMENT: NO MEASURABLE WETLAND DEGRADATION DETECTED

Multi-evidence synthesis does not identify concurrent vegetation and hydrological decline sufficient to establish wetland degradation.

WETLAND EVIDENCE COMPONENT	RESULT
Vegetation degradation (supporting evidence)	0.00%
Surface-water loss (hydrological evidence)	0.00%
Hydro-ecological degradation (concurrent)	0.00%
Mean spectral displacement (CVA)	0.6969

PHYSICAL WETLAND LOSS: NOT ESTABLISHED

Physical wetland loss is not established. In accordance with strict scientific safeguards, optical vegetation change alone is not interpreted as permanent physical wetland loss.

7. EVIDENCE FUSION

Evidence fusion cross-references vegetation transition pathways with surface-water dynamics. A composite hydro-ecological degradation signal requires concurrent evidence of both vegetation stress and surface-water loss, preventing single-sensor optical artifacts from triggering false alarms.

EVIDENCE CLASS	% OF FOOTPRINT	PHYSICAL AREA	DEFINITION & MEANING
Stable landscape	100.00%	35.00 ha	Persistent classified condition across both observation dates
Hydro-ecological degradation	0.00%	0.00 ha	Concurrent vegetation degradation and surface-water loss
Hydro-ecological recovery	0.00%	0.00 ha	Concurrent vegetation recovery and surface-water gain
Vegetation-only degradation	0.00%	0.00 ha	Vegetation decline without concurrent hydrological loss

EVIDENCE FUSION MAP

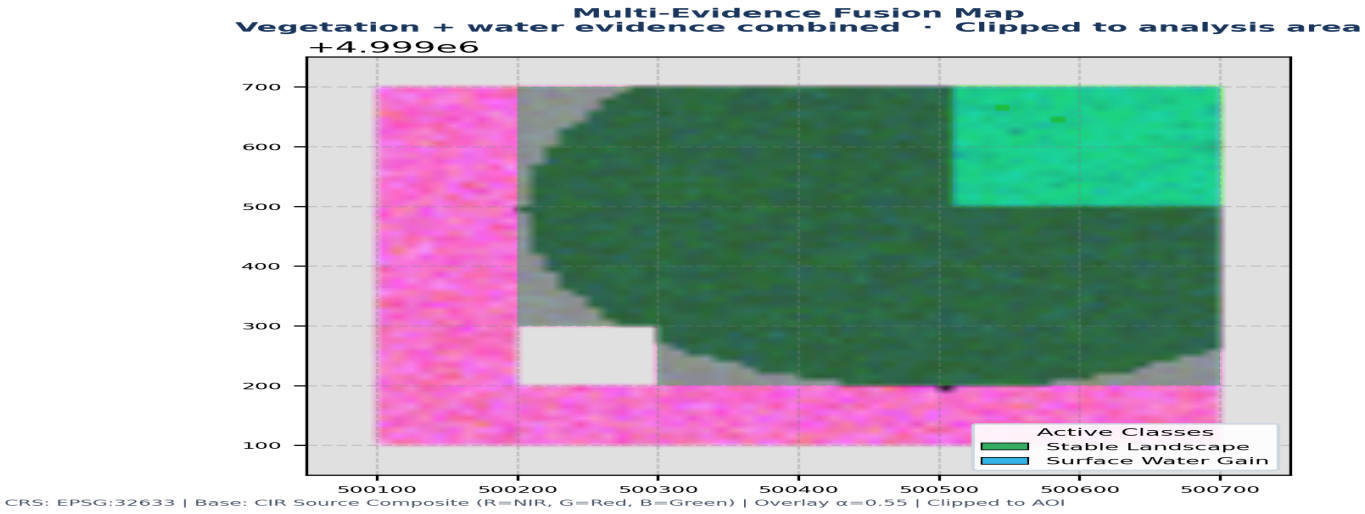


Figure 7: Multi-evidence class map. Hydro-ecological degradation requires concurrent detection of vegetation stress and surface-water loss.

8. DATA QUALITY & SPATIAL ACCOUNTING

Spatial accounting verifies that every spatial component is traceable from source imagery through the inclusion boundary and manual exclusions to the effective analysis footprint.

QUALITY CONTROL CHECK	STATUS	FINDING
Spatial grid alignment	PASS	T1 and T2 rasters verified on a common analysis grid.
Footprint reconciliation	PASS	Selected analysis area reconciled with exclusions (100.00% accounted for).
Independent ground reference	UNAVAILABLE	No external ground-truth validation plots supplied for this run.
Physical depletion guard	ENFORCED	Optical vegetation decline is prevented from being reported as physical wetland loss.
Vegetation class % denominator	FIXED	Class percentages use valid wetland analysis area as denominator, not full raster.

SPATIAL FOOTPRINT AUDIT MAPS

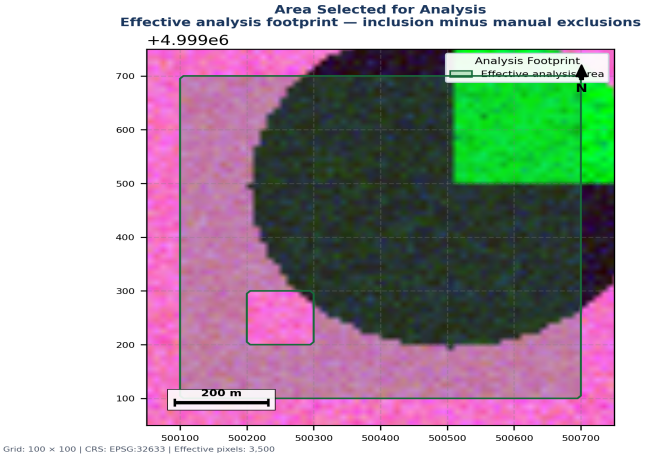


Figure 8a: Effective analysis mask (100.00% active area)

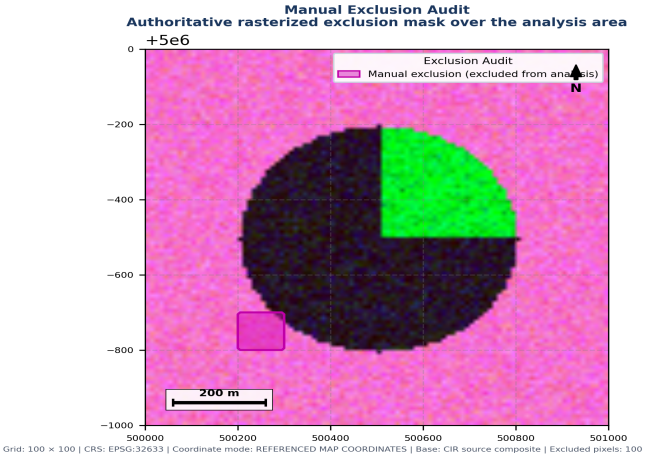


Figure 8b: Exclusion audit map (2.78% removed)

SPATIAL FOOTPRINT RECONCILIATION

FOOTPRINT COMPONENT	% OF SELECTED AREA	PHYSICAL AREA
Original spatial overlap	100.00%	100.00 ha
Selected AOI / Inclusion	100.00%	36.00 ha
Manual exclusion	2.78%	1.00 ha
Effective analysis footprint	100.00% active	35.00 ha

9. TECHNICAL PROVENANCE & LIMITATIONS

Every metric reported in this assessment is derived directly from verifiable raster artifacts produced during the processing pipeline. No analytical values are interpolated or inferred without evidence.

INDICATOR	OBSERVED VALUE	SPATIAL UNIT	PIPELINE ARTIFACT / METHOD
Wetland source	PROJECT_BOUNDARY	class	Wetland mask source selected by wetland context stage
Wetland context status	WETLAND_BOUNDARY_AVAILABLE	status	Wetland mask evaluation result
Effective analysis area	100.00%	active footprint proportion	alignment/effective_analysis_mask.tif
Manual excluded area	2.78%	footprint proportion	exclusions/manual_exclusion_mask.tif
Vegetation degradation	0.00%	% of effective footprint	degradation/degradation_map.tif (H→M, H→S, M→S)
Vegetation recovery	0.00%	% of effective footprint	degradation/recovery_map.tif (M→H, S→M, S→H)
Veg class proportion basis	Valid wetland analysis area	area proportion	metrics.class_area_metrics — wetland-masked extent, not full raster
CVA Mean Magnitude	0.6969	index units	change/cva_magnitude.tif (||[ΔNDVI, ΔGNDVI, ΔNDRE]||)
Mean ΔNDVI	0.4944	index units	change/delta_ndvi.tif (NDVI_T2 - NDVI_T1)
NDWI Water Threshold	0.00	index units	Calibrated Green/NIR NDWI threshold
Physical depletion assessment	NOT_ASSESSED	status	Independent-evidence claim guard

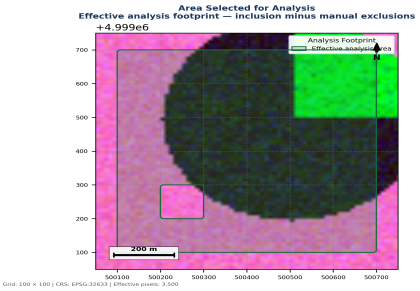
SCIENTIFIC LIMITATIONS & BOUNDARIES

Observation Interval: Remote-sensing comparisons reflect conditions at specific acquisition times. Short-term intervals capture immediate physiological or seasonal responses that may differ from multi-year ecological trends.
Model Inferences: Machine learning vegetation classifications are probabilistic estimates. In-situ field validation is recommended before using findings for regulatory, legal, or compensation purposes.
Spatial Reference: Where imagery lacks a validated Coordinate Reference System (CRS), proportional footprint metrics serve as the authoritative spatial unit. Physical area (hectares) is reported only when georeferencing is confirmed.
Percentage Accounting: Vegetation class percentages are computed relative to the valid wetland analysis area within the effective footprint. Transition and hydrological percentages are computed relative to the effective analysis footprint (35.00 ha). No individual category proportion should exceed 100%.

Run: run_20260922_082400 · **Software:** GIS Engine v1.0.0 · **Analysis Basis:** RASTER_CRS_AND_TRANSFORM

REPRODUCIBILITY & DATA INTEGRITY AUDIT

AUDIT DIMENSION	VERIFICATION STATUS	AUDIT RESULT
Pipeline execution	DETERMINISTIC	Fully automated, artifact-driven pipeline with zero manual interpolation.
Raster co-registration	VERIFIED	Common spatial coordinate space (EPSG:32633).
Spatial accounting	RECONCILED	Inclusion, exclusion, and effective footprint sum to 100.00%.
Multi-evidence fusion	CONFIRMED	Vegetation, spectral, and hydrological streams independently verified.
Reporting claim integrity	COMPLIANT	No physical wetland loss asserted without concurrent hydrological evidence.



Spatial audit domain (EPSG:32633)